

CDR1 and CDR2 encoded by germline in V
 CDR3 by DJ and junctional for VDJ B and J + VJ junctional for a

α chain
Chr 14

V	J		C
n=~45	n=~50		

β chain
chr 7

V	D	J	C
n=~50	n=2	n=12	

junctional diversity
 (non-germline)

largest contributor to diversity

P nucleotides (template after hairpin opening by Artemis)

N nucleotides (then nontemplate followed by DNA ligation) (20 by TdT) - more common in B chain

10^{15} - 10^{20} unique TCR combinations

combinatorial diversity
 (germline)

1 to 3 million upper limit)